Two Samples, One Difference

AP Statistics · Unit 4 · Topic 4.6 · B: 2027-01-22 · E: 2027-03-10 · TEACHER KEY

CED TETHER

- **ENDURING UNDERSTANDING: UNC-3** — Probabilistic reasoning allows us to anticipate patterns in data.
- **Skills: 3.B** — Determine parameters for probability distributions. | 3.C — Describe probability distributions. | 4.B — Interpret statistical calculations and findings to assign meaning or assess a claim.
- UNC-3.T Determine parameters of a sampling distribution for a difference in sample means. [Skill 3.B]
- UNC-3.T.1 For a numerical variable, when randomly sampling from two independent populations with means μ1 and μ2 and standard deviations σ1 and σ2, the sampling distribution of the difference in sample means has mean μ = μ1 - μ2 and standard deviation σ = sqrt($\sigma 1^2$/n1 + $\sigma 2^2$/n2). UNC-3.T.2 If sampling without replacement and sample sizes are less than 10% of population sizes, the difference is negligible.
- UNC-3.U Determine whether a sampling distribution of a difference in sample means can be described as approximately normal. [Skill 3.C]
- UNC-3.U.1 The sampling distribution of the difference in sample means can be modeled with a normal distribution if both population distributions can be modeled with normal distributions. UNC-3.U.2 If population distributions cannot be modeled with normal distributions, the sampling distribution can be approximated by a normal distribution if both sample sizes are at least 30.
- UNC-3.V Interpret probabilities and parameters for a sampling distribution for a difference in sample means. [Skill 4.B]
- UNC-3.V.1 Probabilities and parameters for a sampling distribution for a difference of sample means should be interpreted using appropriate units and within the context of specific populations.

Essential question: How can two sample means use different pathways to justify one normal model for their difference?

DOK-3 skill: 4.B · **FRQ pattern:** `two-shape-routes-for-a-mean-difference`

DOK rationale: Part (c) coordinates a separate shape pathway for each sample mean, judges two incomplete model claims, and uses a small-sample counterfactual to trace when the tail probability loses support.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Ask which details govern the sampling model.
- Begin the lesson video follow-along at minute 5.
- Return at minute 33. Require separate shape arguments and the counterfactual check.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Name one argument to retain and one that needs more evidence.
- Complete the follow-along, finish (a)–(c), revise, and turn in.

Questions to ask

- Which distribution are we justifying for North: individual waits or sample means?
- Why can the two groups use different routes to approximate normality?
- What changes if the skewed South sample has size 8?
- What repeated-sampling event does 0.0273 describe?

Adult role
Solo teacher. Justify North and South separately, then combine them using independence.

[WARN] WATCH FOR

- Requiring both samples to have size at least 30 even when one population is normal.
- Assuming one normal component automatically fixes an unjustified second component.
- Subtracting variances because the statistic subtracts means.

SCORING PART (c) — E / P / I

Expected elements

- **uses-north-normal-route** (required) — Uses North population normality despite $n_N = 12$
- **uses-south-clt-route** (required) — Uses South $n_S = 48$ with the CLT
- **coordinates-difference-model** (required) — Uses both component models and independence for 0.0273
- **adjudicates-both-accounts** (required) — Diagnoses both incomplete rules
- **traces-counterfactual** (required) — Uses $n_S = 8$ to expose the combined-size shortcut

E	Justifies both components by distinct routes, coordinates independence, diagnoses both accounts, and explains the $n_S = 8$ consequence.
P	Reaches the warranted verdict but omits a route, independence, or the counterfactual consequence.
I	Requires both populations normal or uses one/combined sample size to justify both means.

[STAR] TODAY'S PROBLEM — annotated

Suppose parcel pickup waits have the shown population parameters. North waits are approximately normal; South waits are strongly right-skewed. Independent SRSs are taken without replacement, and the service wants $P(\bar{X}_N - \bar{X}_S > 2.5)$.

Rina: "South is skewed, so I would not use a normal model for the difference."

Oscar: "North is normal and the combined sample size is 60, so the normal model is valid; the probability is 0.0273."

Locker samples

Locker	n	μ	σ
North	12	6.0	1.8
South	48	4.8	3.0

FIRST TAKE (5 min)

Identify one argument you would retain and one that needs more evidence. Cite a population shape or sample size.
Key: Not graded. Each account contains a claim that still needs component-level justification.

Rules callout "JUSTIFY EACH SAMPLE MEAN BEFORE THEIR DIFFERENCE (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** State the mean of $\bar{X}_N - \bar{X}_S$ and translate the event above into context.

Key: The sampling distribution is centered at $\mu_N - \mu_S = 6.0 - 4.8 = 1.2$ minutes. The event says the sample mean North wait exceeds the sample mean South wait by more than 2.5 minutes.

DOK 2 **(b)** Verify both 10 percent conditions, compute the sampling-distribution SD, and calculate the probability under a normal model.

Key: North: $12 \le 0.10(600) = 60$; South: $48 \le 0.10(1{,}000) = 100$. The SD is $\sqrt{1.8^2/12 + 3.0^2/48} = \sqrt{0.4575} = 0.676387$. Thus $z = (2.5 - 1.2)/0.676387 = 1.92198$ and the upper-tail probability is $0.027304 \approx 0.0273$.

DOK 3 **(c)** Adjudicate both accounts and decide whether 0.0273 is warranted. Justify each sample mean separately using its shape and sample size, coordinate independence, then use a South sample of 8 to expose the consequence of any rejected shortcut.

Key: The probability is warranted, but both explanations are incomplete. North's normal population supports $\bar{X}_N$ with $n_N = 12$. For skewed South, $n_S = 48 \ge 30$ supports $\bar{X}_S$ by the CLT. Independence then supports an approximately normal difference. Rina would discard a justified probability by requiring both populations to be normal. Oscar's combined-size rule fails because each component needs support: with skewed South and $n_S = 8$, its sample mean is not assured normal, so neither is the difference. That shortcut would make readers trust unsupported tails.

EXIT

One thing the video changed about my first take: ______________________________